PROBLEM 1.2

Separable ODE • Asymptotics • Doubling Time

Malthusian Population Dynamics & Doubling Time

Original statement · ProblemsCh1.pdf

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Let $x(t)$ be the population of a certain species as a function of time, and suppose that the rate of change of this population is proportional to the existing population at each time, with constant of proportionality $k \in \mathbb{R}$.

(i) Write a differential equation for $x(t)$.
(ii) Solve it, determine the behavior of $x(t)$ as $t \to \infty$ and interpret it depending on the sign of k.
(iii) If the initial population is $x(0) = x_0$ and $k > 0$, determine the time needed for the population to double.

PHASE 2: FUNDAMENTAL FORMULATION

The rate of change is the time derivative $\frac{dx}{dt}$. Proportionality to $x(t)$ with constant k yields:

$$\frac{dx}{dt} = kx(t), \quad x(0) = x_0 > 0$$

This is a first-order linear separable homogeneous ODE.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): By definition, the ODE is:

$$\mathbf{\frac{dx}{dt} = kx} \tag{1.2.1}$$

Part (ii): Separate variables assuming $x > 0$:

$$\frac{1}{x}\,dx = k\,dt$$

Integrate both sides from initial state $(0, x_0)$ to arbitrary state $(t, x(t))$:

$$\int_{x_0}^{x(t)} \frac{1}{\xi}\,d\xi = \int_0^t k\,ds \implies [\ln \xi]_{x_0}^{x(t)} = k[s]_0^t \implies \ln\left(\frac{x(t)}{x_0}\right) = kt$$

Exponentiate both sides:

$$\frac{x(t)}{x_0} = e^{kt} \implies \mathbf{x(t) = x_0 e^{kt}} \tag{1.2.2}$$

Asymptotic limits as $t \to \infty$:

- If $k > 0$: $\lim_{t\to\infty} x(t) = \lim_{t\to\infty} x_0 e^{kt} = +\infty$ (Exponential Malthusian explosion; natality exceeds mortality).
- If $k = 0$: $x(t) = x_0 e^0 = x_0$ (Static equilibrium; births perfectly balance deaths).
- If $k < 0$: $\lim_{t\to\infty} x(t) = \lim_{t\to\infty} x_0 e^{-|k|t} = 0$ (Exponential extinction; mortality exceeds natality).

Part (iii): Doubling Time t_d:

Set $x(t_d) = 2x_0$ with $k > 0$:

$$x_0 e^{kt_d} = 2x_0 \implies e^{kt_d} = 2$$

Take the natural logarithm:

$$kt_d = \ln 2 \implies \mathbf{t_d = \frac{\ln 2}{k}} \tag{1.2.3}$$

PHASE 4: FINAL RESULT & DIMENSIONAL VERIFICATION

BOXED RESULT

$$x(t) = x_0 e^{kt}, \quad t_d = \frac{\ln 2}{k}$$

Dimensional Verification: $[k] = \text{time}^{-1} \implies [t_d] = \frac{1}{[k]} = \text{time}$, which is dimensionally homogeneous.